👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

**UNIT 6 · STANDARD 6.NOS.4**

# Least Common Multiple

Lesson 6-12 · Teacher Copy

**Name:** _____ **Date:** _____ **Class:** _____

## My Notes

**Level 1 Support**

🎯 **TODAY'S OBJECTIVES**

**Content Objective: I can find the least common multiple (LCM) of two numbers by listing or comparing their multiples.**

**Language Objective: I can explain how I found the LCM using the words multiple, common multiple, skip counting, and least common multiple.**

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

**Least Common Multiple** | **Multiple** | **Common multiple** | **Skip counting** | **Prime factorization**

**1** The smallest number that is a multiple of both is their _____ _____ _____.

**2** The product of a number and any whole number is a _____ of that number.

3. A multiple that two or more numbers share is a _____ multiple.

4. Counting forward by a number other than 1, like 6, 12, 18, is called _____.

5. Writing a number as _____ numbers multiplied together is its prime factorization.

## Write About the Math

**How does the LCM help you figure out the fewest hot dogs and buns to buy so they come out even?**
*¿Cómo te ayuda el mínimo común múltiplo a averiguar la menor cantidad de salchichas y panes que hay que comprar para que salgan parejos?*

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Least Common Multiple** — *Mínimo común múltiplo*
- ☐ **Multiple** — *Múltiplo*
- ☐ **Common multiple** — *Múltiplo común*
- ☐ **Skip counting** — *Conteo salteado*
- ☐ **Prime factorization** — *Factorización prima*

**Model:** *The next shared check is a common multiple of 4 and 6, not their sum.* — Students reason that each astronaut repeats on multiples of their own number, so they only overlap at a common multiple, and adding 4 + 6 does not land on both schedules.

## Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **I need to buy ___ hot dogs and ___ buns because the LCM of 8 and 6 is ___.**
- **My answer is ___ because ___.**
  *Mi respuesta es ___ porque ___.*

## Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is ___.**
  *Mi afirmación es ___.*
- **My evidence is ___, so ___.**
  *Mi evidencia es ___, entonces ___.*

## Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

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## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Least Common Multiple
2. **Notes 2:** Multiple
3. **Notes 3:** Common multiple
4. **Notes 4:** Skip counting
5. **Notes 5:** Prime factorization
6. **Watch for this mistake:** *A common mistake in Least Common Multiple is multiplying the two numbers together instead of listing multiples to find the smallest shared one. For example, for 4 and 6, a student calculates 4 × 6 = 24 and calls that the LCM, but the actual least common multiple is 12 (multiples of 4: 4, 8, 12; multiples of 6: 6, 12 — 12 is the first number that appears in both lists). This shortcut only works when the two numbers share no common factors (like 4 and 9, where the LCM really is 36 = 4 × 9); for numbers like 4 and 6 that share a factor of 2, multiplying overshoots the true LCM. Always list multiples of both numbers and pick the first one that appears in both lists.*
7. **Try It 1:** C. 12 — *Skip count: multiples of 4 are 4, 8, 12; multiples of 6 are 6, 12. The first shared multiple is LCM(4, 6) = 12 minutes.*
8. **Try It 2:** C. 24 — *LCM(8, 12) = 24. Multiples of 8: 8, 16, 24; multiples of 12: 12, 24. The first shared multiple is 24 minutes.*